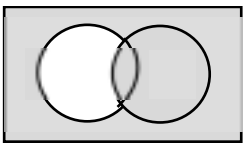
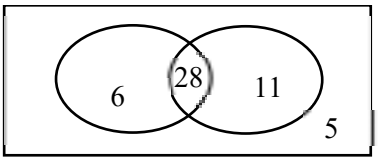
Question	Answer	Marks	Partial Marks
5(c)	78.2 or 78.17 to 78.19…	**3**	**M2** for $\frac{1}{2}\times 10.6\times(6.4+\textit{their } 8.68)\times\sin(78)$ oe OR **M1** for $\frac{1}{2}\times 10.6\times 6.4\times\sin(180-78)$ oe **M1** for $\frac{1}{2}\times 10.6\times\textit{their } 8.68\times\sin 78$ oe
6(a)		**1**	
6(b)		**2**	**B1** for 2 or 3 correct elements or **M1** for $34-x$, x and $39-x$ correctly placed on diagram and $x = 28$
6(c)(i)	8	**1**	
6(c)(ii)	11	**1**	
6(c)(iii)	2	**1**	
6(c)(iv)	$C\cap S\cap B'$ oe	**1**	
6(c)(v)	$\frac{19}{30}$ oe	**1**	
6(c)(vi)	$\frac{2}{57}$ oe	**3**	**M2** for $\frac{4}{19}\times\frac{3}{18}$ or **M1** for $\frac{4}{19}$ seen
6(c)(vii)	Equal numbers 15 or equal probability $\frac{15}{30}$ oe	**1**	
7(a)	$\frac{x+5}{x+4}$ final answer	**3**	**B1** for $(x-5)(x+5)$ **B1** for $(x-5)(x+4)$
7(b)	$\frac{2x^2+12x-5}{x(x-1)}$ or $\frac{2x^2+12x-5}{x^2-x}$ final answer	**3**	**B1** for common denominator $x(x-1)$ oe **B1** for $(x-1)(x+5)+x(x+8)$ or better
7(c)(i)	$6x^2-8x$ final answer	**2**	**B1** for each term in final answer or **M1** for correct answer seen and spoilt

Question	Answer	Marks	Partial Marks
7(c)(ii)	64	**2**	**FT** *their* **(c)(i)** correctly evaluated provided at least 2 terms but not the original equation **M1** for substituting $x = 4$ into *their* **(c)(i)**
7(c)(iii)	(0, 6) $\left(\frac{4}{3}, \frac{98}{27}\right)$ oe	**4**	**M1** for *their* derivative = 0 or $\frac{dy}{dx} = 0$ soi **B1** for $x = 0$ and $x = \frac{4}{3}$ **M1dep** for substituting one of *their x* values into $y = 2x^3 - 4x^2 + 6$ soi
8(a)(i)	3 22 43 48 50	**2**	**B1** for 4 correct or **M1** for one error in adding.
8(a)(ii)	correct diagram	**3**	**B1FT** *their* **(a)(i)** for 5 correct heights **B1** for 5 points at upper ends of intervals on correct vertical line **B1FT dep on at least B1** for increasing curve through *their 5* points After 0 scored, **SC1** for 4 of *their* points correctly plotted
8(a)(iii)	35 to 38	**1**	
8(b)	Correct box-and-whisker diagram 1.45 1.57 1.64 1.71 1.83	**4**	**B1** for median 1.64 drawn **B1** for LQ 1.57 drawn **B1** for UQ 1.71 drawn If 0 scored **SC1** for 1.64, 1,57 or 1.71 seen
9(a)	1350 or 1354….	**6**	**M2** for $20^2 - 13^2$ or **M1** for $BC^2 + 13^2 = 20^2$ **A1** for $\sqrt{231}$ or 15.2 or 15.19 to 15.20 **M1** for 20×24 and 13×24 and *their* 15.2×24 **M1** for [½ ×] *their* 15.2×13
9(b)	2370 or 2369 to 2371… cao	**1**	
9(c)	24.6 or 24.58 to 24.59	**4**	**M3** for sin [...] = $\frac{13}{\sqrt{20^2 + 24^2}}$ oe or **M2** for $\sqrt{20^2 + 24^2}$ or $\sqrt{24^2 + 20^2 - 13^2}$ or **M1** for $AF^2 = 20^2 + 24^2$ or $24^2 + 20^2 - 13^2$ or **M1** for correct angle identified
10(a)	0.75 3 7 3 0.75	**3**	**B2** for 4 or 3 correct or **B1** for 2 correct